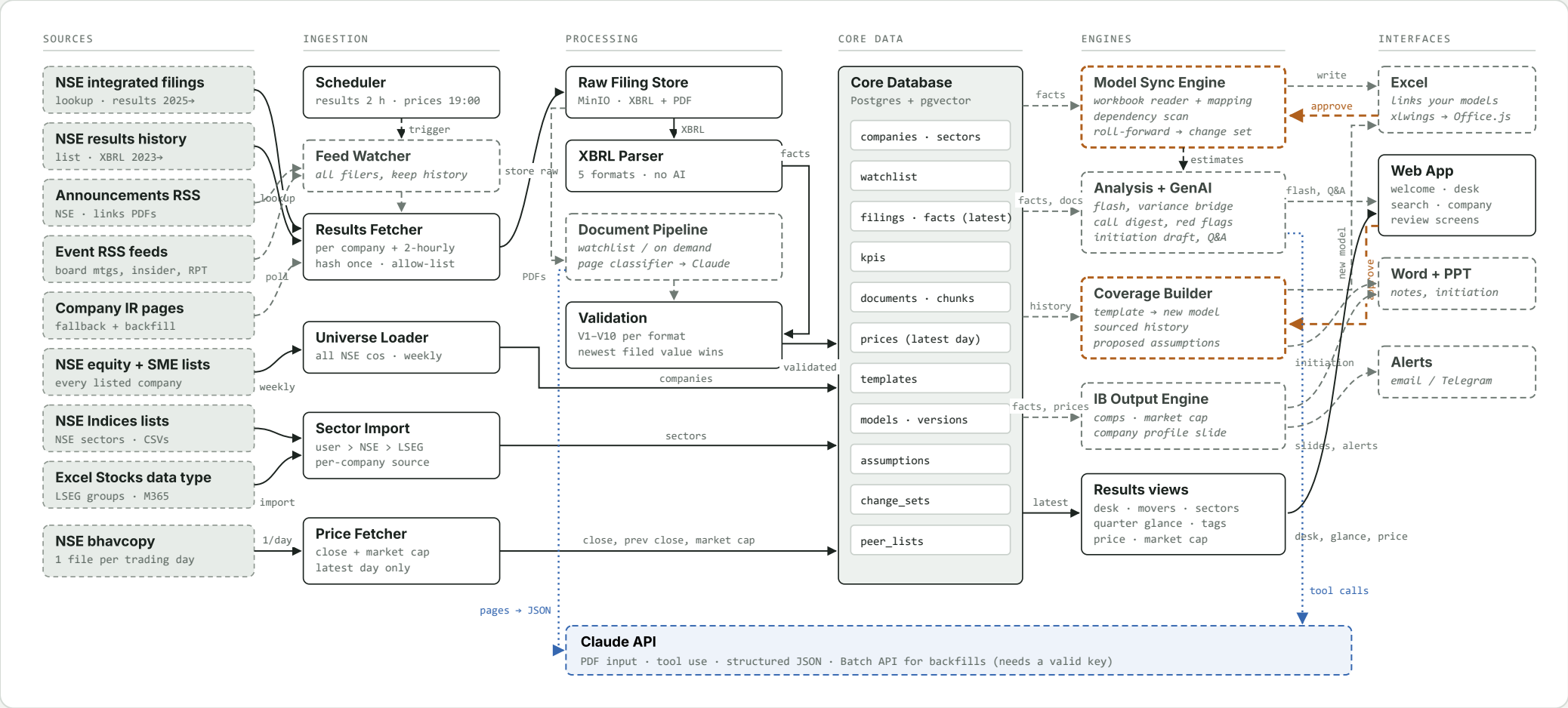
SYSTEM DESIGN V0.0.4 · 18 OCTOBER 2026 · PHASE 0, PHASE 1 UNDER WAY

Analekh Architecture

The analyst's record, kept for you.

Every NSE-listed company is preloaded. Its results are fetched from NSE's integrated filings and, back to March 2023, its financial-results list, then parsed and validated on both bases for every filer. Closing prices come once a day from NSE's bhavcopy. For a company you follow, the system will find and update your Excel model after you approve each change, or build a new model from your sector template and draft an initiation. Core numbers come from XBRL; the AI writes the analysis and pulls every figure from the database.

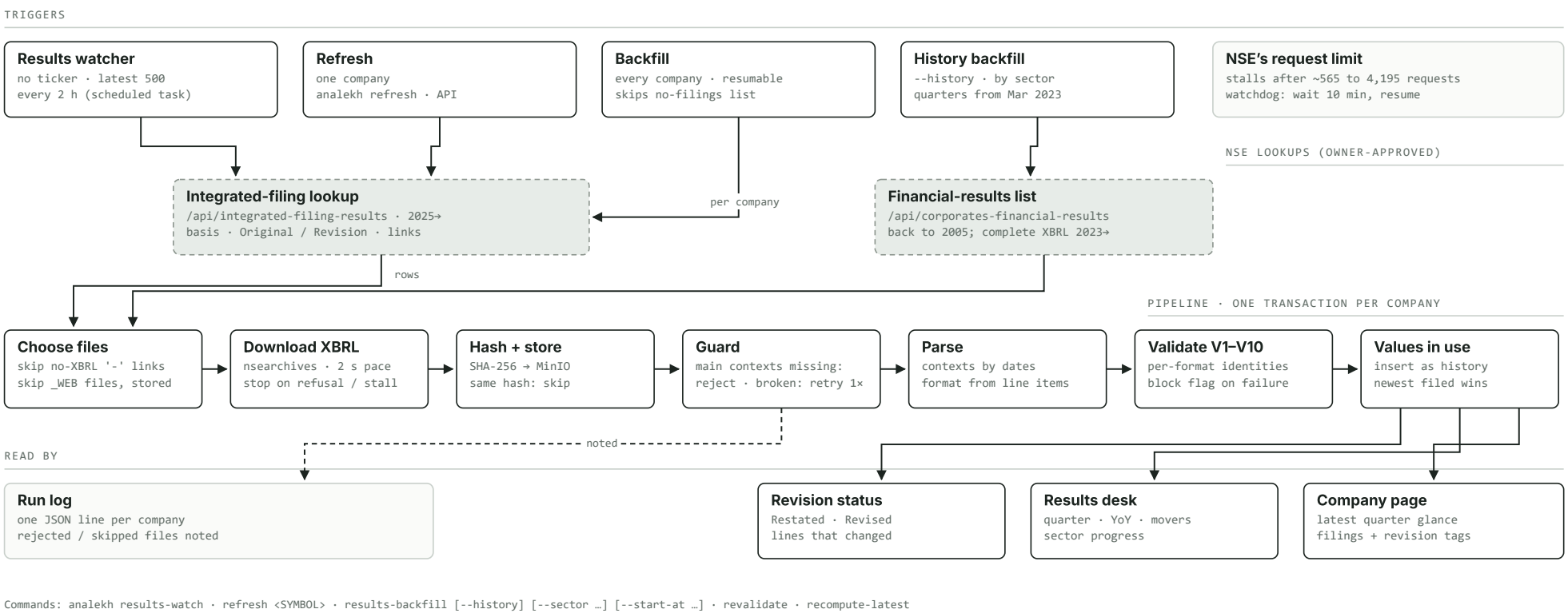
Components and data flow



data · control / return · analyst approval · Claude API call · external source · built, running on real data · core built or planned, not running yet

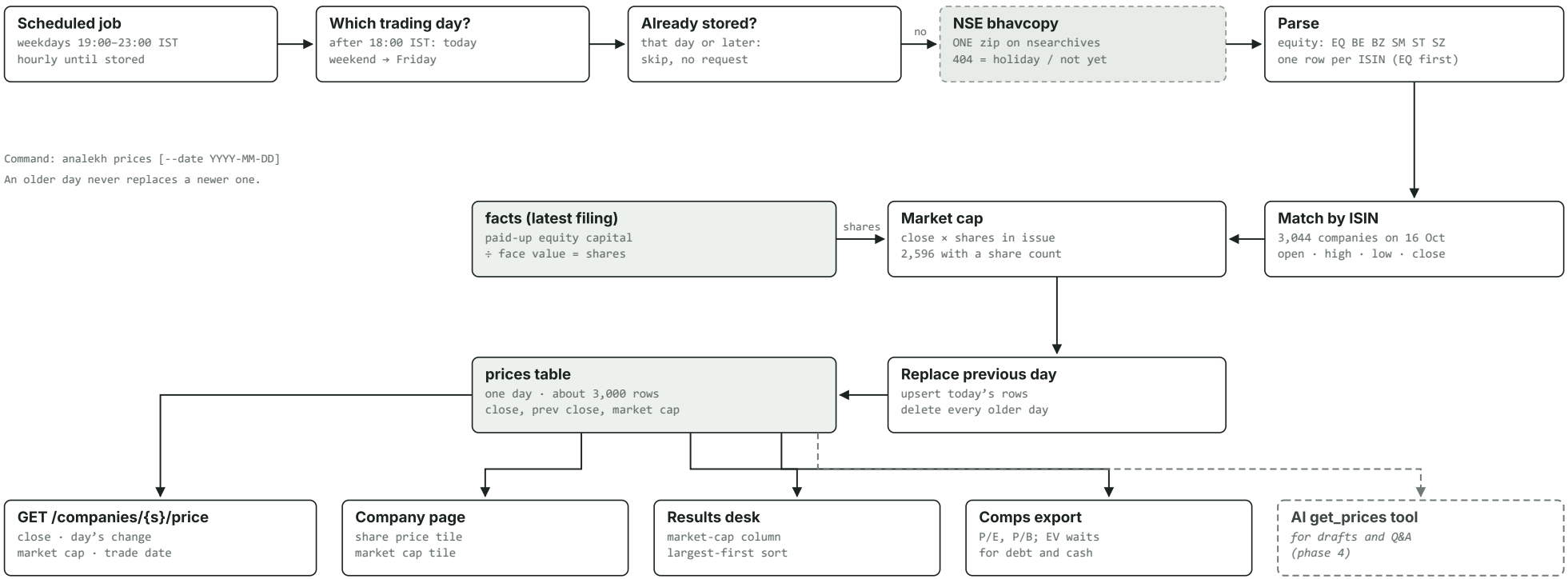
Results come from NSE's integrated-filing lookup (per company, and one request every 2 hours without a ticker for the latest filings) and its financial-results list for 2023–2024. When a quarter is refiled, the most recently filed values are used and the older ones kept as history. Sectors follow NSE's own taxonomy first. Prices: one bhavcopy per trading day; only the latest close is kept. Only the relevant PDF pages will reach Claude, and nothing is written to Excel until the analyst approves.

Workflow 1 · Results collection BUILT



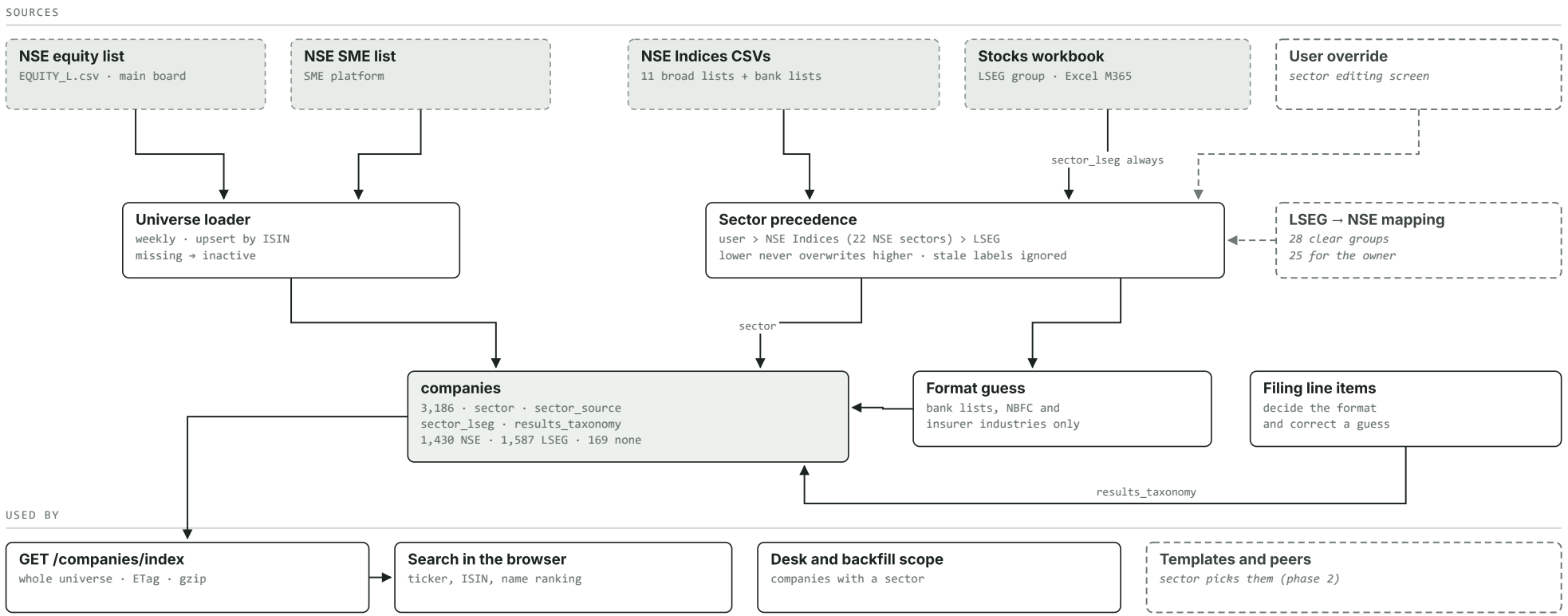
Four triggers, two NSE lookups, one pipeline. The watcher asks once, without a ticker, for the latest filings across all companies; a refresh or backfill asks per company. The history backfill reads the financial-results list, whose files are complete from March 2023. Each file is downloaded once (SHA-256, stored in MinIO); files missing their main contexts are rejected and a broken file is retried once, then skipped, so one bad file never costs a company its other filings. Values are inserted as history and the newest filed value of each line becomes the one in use, whatever order files arrived in. One transaction per company: a stopped run keeps what it stored and resumes at the company it stopped on. NSE pauses after a few hundred to a few thousand requests; runs wait 10 minutes and go on.

Workflow 2 · Daily closing prices BUILT



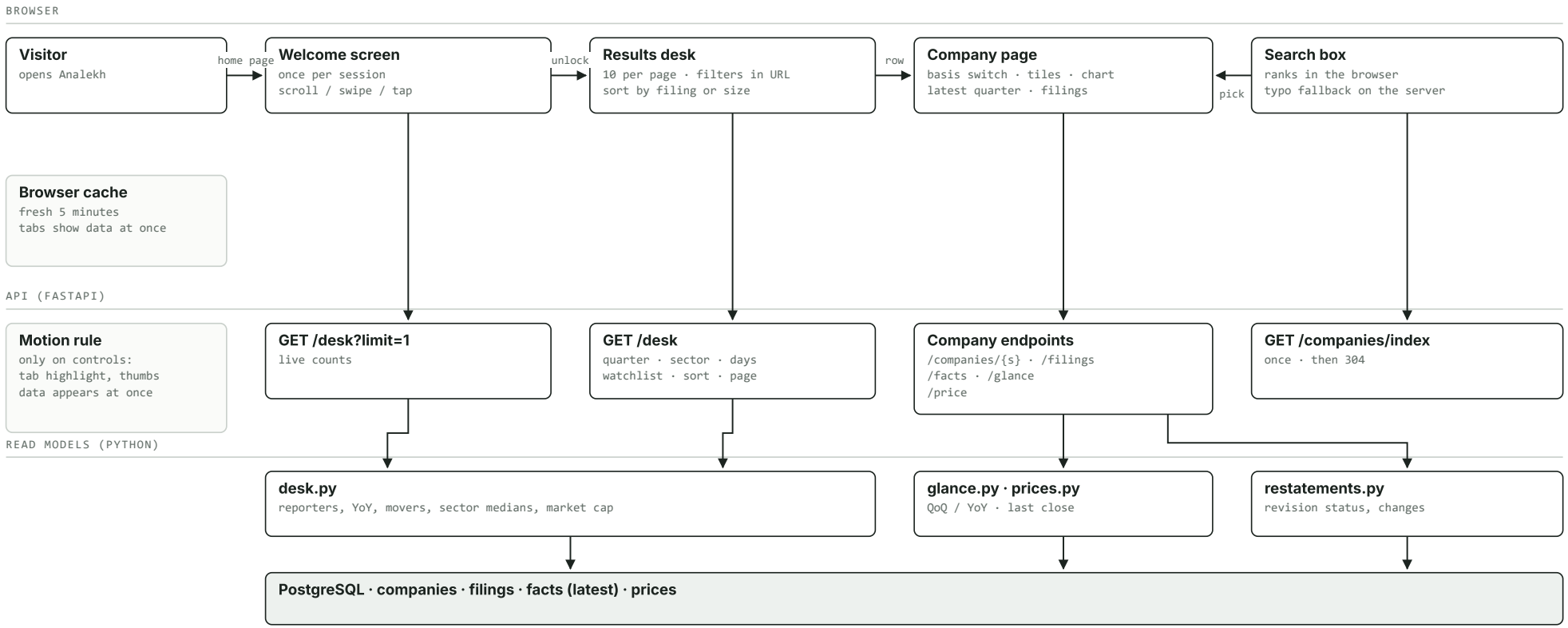
One file per trading day, after the close. NSE publishes the bhavcopy in the evening; the job tries hourly from 19:00 to 23:00 IST on weekdays until the day is stored, then does nothing. Market cap is the close times shares in issue, worked out from the latest results filing (paid-up equity capital divided by face value). Only the latest close is kept (owner's rule, 18 October 2026): each file replaces the previous day's prices, and the day's change comes from the previous close the file carries. No live prices, no price history.

Workflow 3 · Company list and sectors BUILT



Every NSE company, then its sector. The equity and SME lists are loaded weekly. Sectors follow NSE's own taxonomy (22 sectors) from NSE Indices' published constituent lists, which outrank the LSEG industry groups imported from an Excel Stocks data type workbook; a user's value will outrank both. The LSEG group is always kept in its own column. The results format (Ind-AS, bank, NBFC, general or life insurer) is first guessed from strong signals only, then decided by the line items each filing tags.

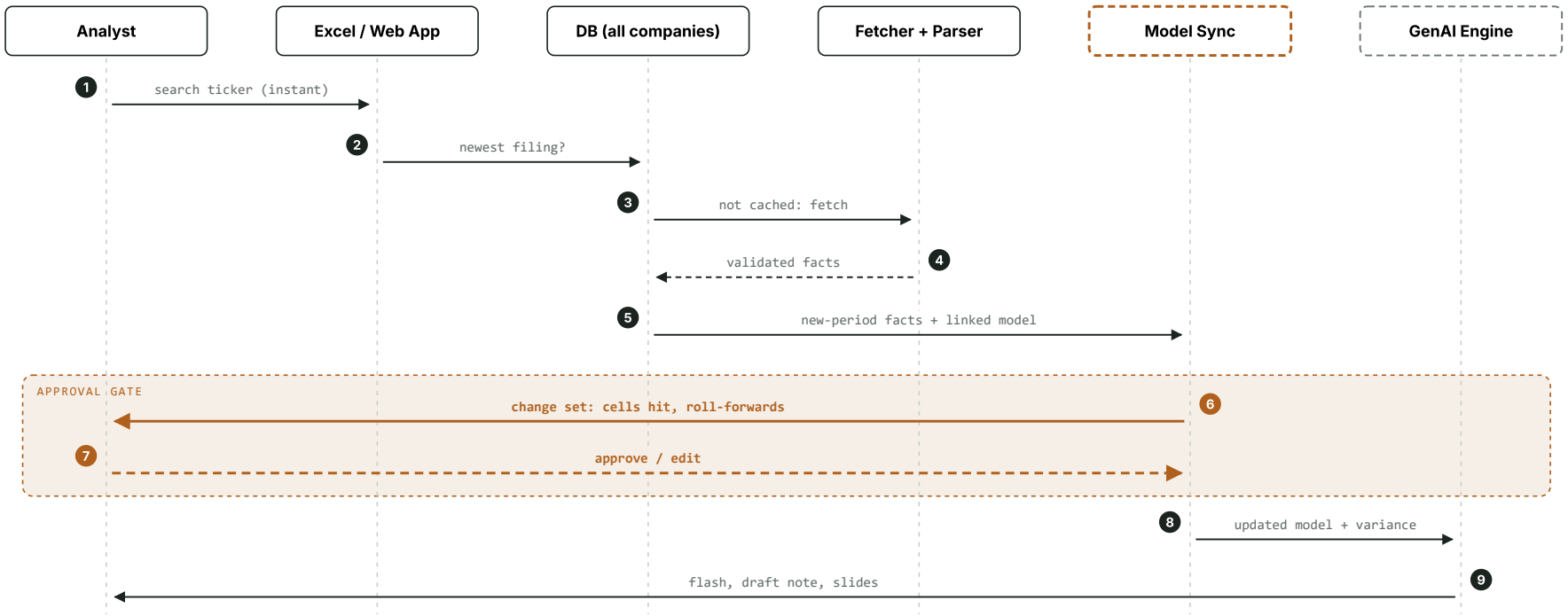
Workflow 4 · Opening Analekh and reading results BUILT



Welcome, then the desk. The first visit of a browser session to the home page shows the welcome screen with live counts; a scroll, swipe or tap unlocks it into the Results desk. Search runs in the browser over the whole universe. Data stays fresh in the browser for five minutes, so moving between tabs shows it at once. Motion is only on what you click (the sliding tab highlight and switch thumbs); figures and screens appear immediately.

Workflow 5 · Quarterly model update

PLANNED · PHASE 2



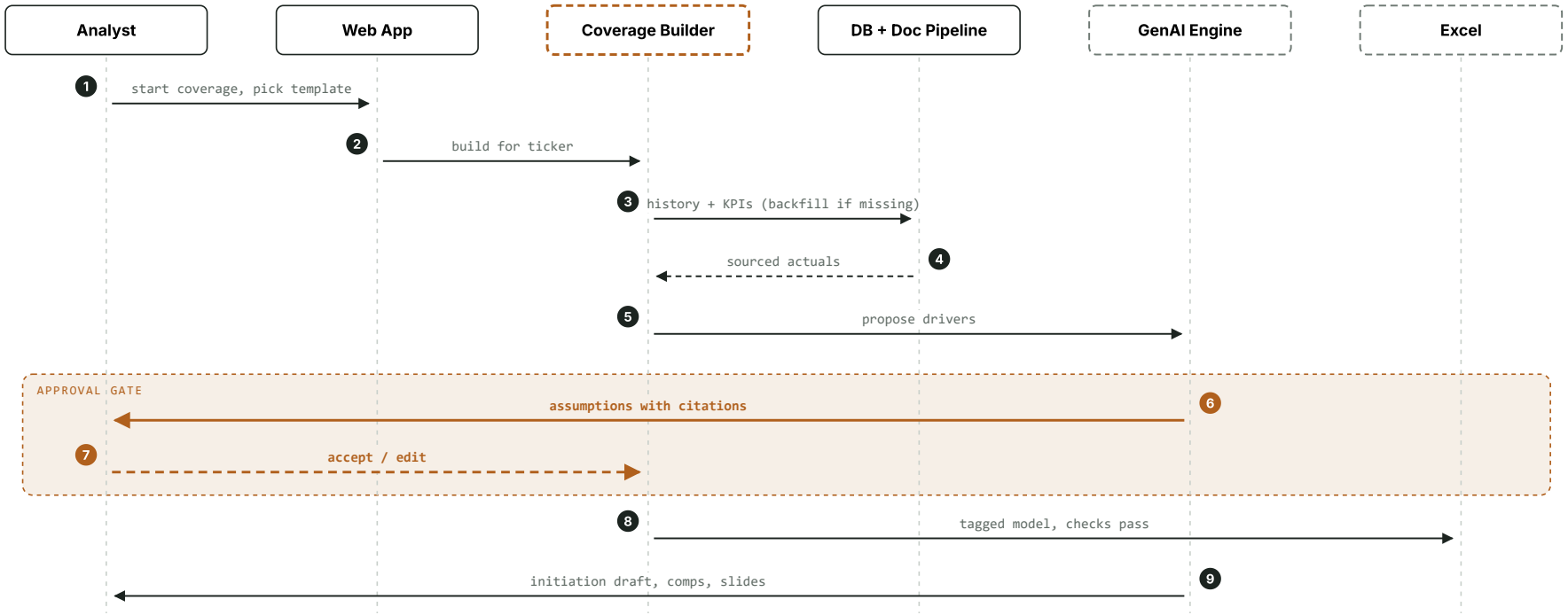
Steps 3 and 4 run only when the filing isn't already in the database (for most companies it is, collected by workflow 1). The linked model is found through watched folders or the open workbook, identified by a hidden tag, its file name or its contents. Steps 6 and 7 are the approval gate: the analyst sees which cells change and which forecasts need rolling forward, accepts or edits each one, and only then is a new model version saved. The original workbook is never overwritten.

What changed in v0.0.4

- **History:** NSE's financial-results list (owner-approved) adds 2023–2024; about 15 quarters per company.
- **Prices:** NSE's bhavcopy once a day; market cap from the filing's share count; only the latest close kept.
- **Sectors:** NSE's own taxonomy first; LSEG groups kept apart.
- **Watcher:** every 2 hours now; a SEBI-season schedule proposed, backed by NSE's timings.
- **Web app:** welcome screen, prices and market cap, gliding tabs and switches.
- **Future scope:** a live earnings-call listener, ahead of the filed recording.

Workflow 6 · Start coverage

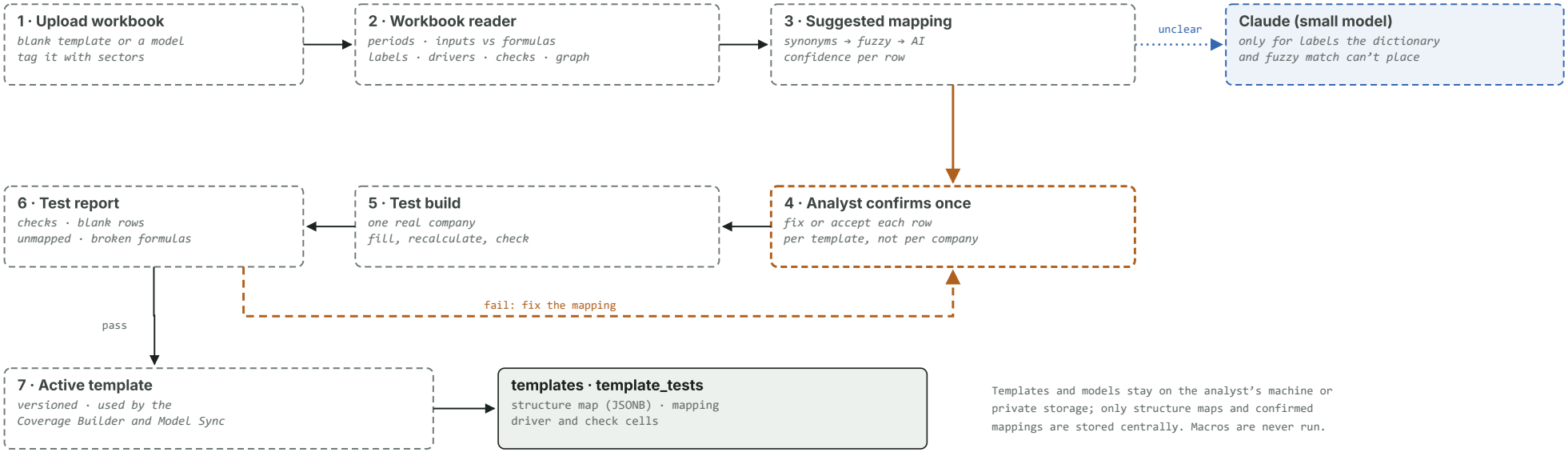
PLANNED · PHASE 2



Used when no model exists yet. The analyst picks a company and a sector template (their own upload or a built-in one). The builder fills the template's inputs with sourced history and keeps its formulas. Proposed assumptions pass through the same approval gate before the tagged model is saved. From the next results onward, the quarterly update takes over.

Workflow 7 · Template onboarding

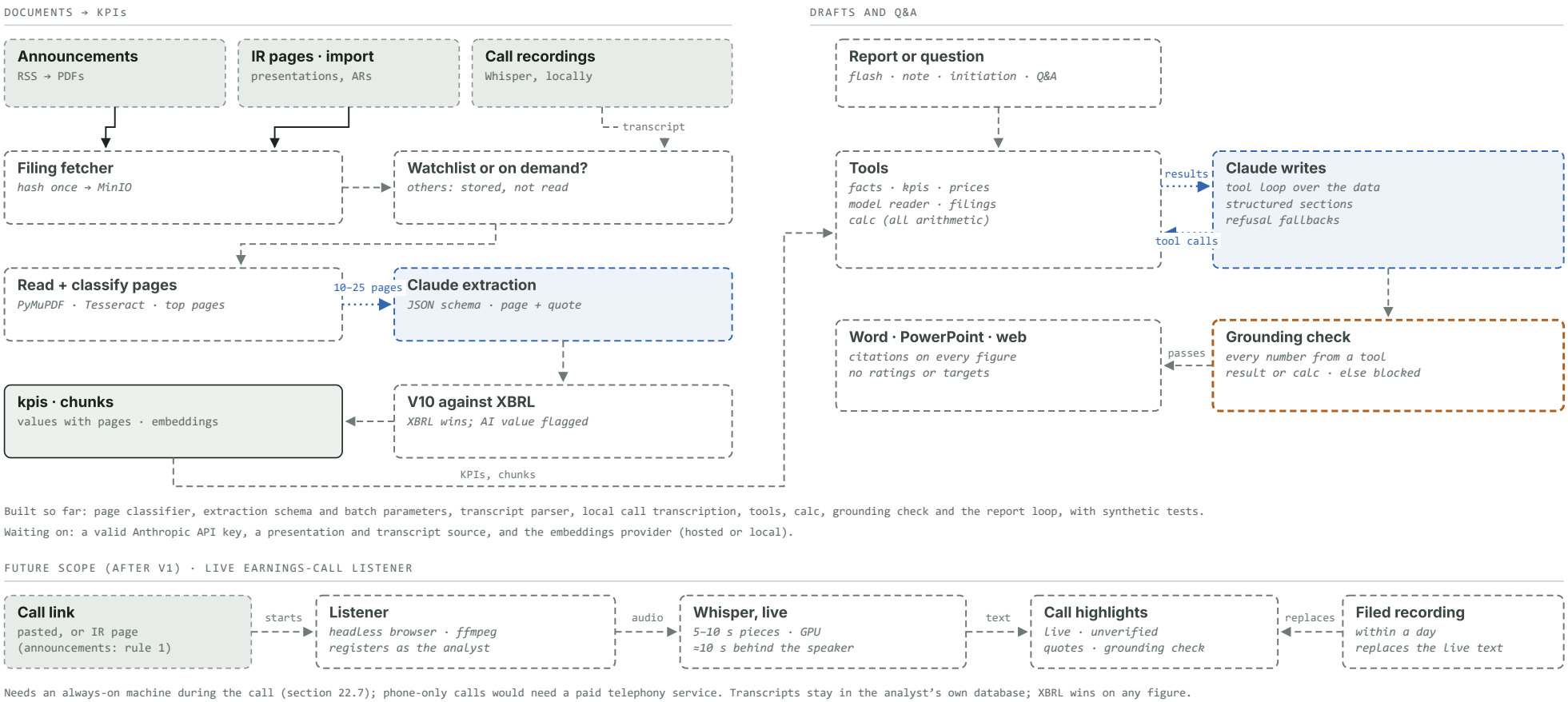
PLANNED · PHASE 2



Confirm once per template. The workbook understanding engine reads any workbook: its period axis, inputs versus formulas, row labels, drivers, checks and the formula graph. It suggests where each standard line goes, with a confidence for each row; the analyst confirms or fixes it once. A test build with a real company must pass (checks hold, no broken formulas) before the template can be used.

Workflow 8 · Documents, KPIs and grounded drafts

PLANNED · PHASES 3–4 · LIVE CALLS: FUTURE SCOPE



Built so far: page classifier, extraction schema and batch parameters, transcript parser, local call transcription, tools, calc, grounding check and the report loop, with synthetic tests.

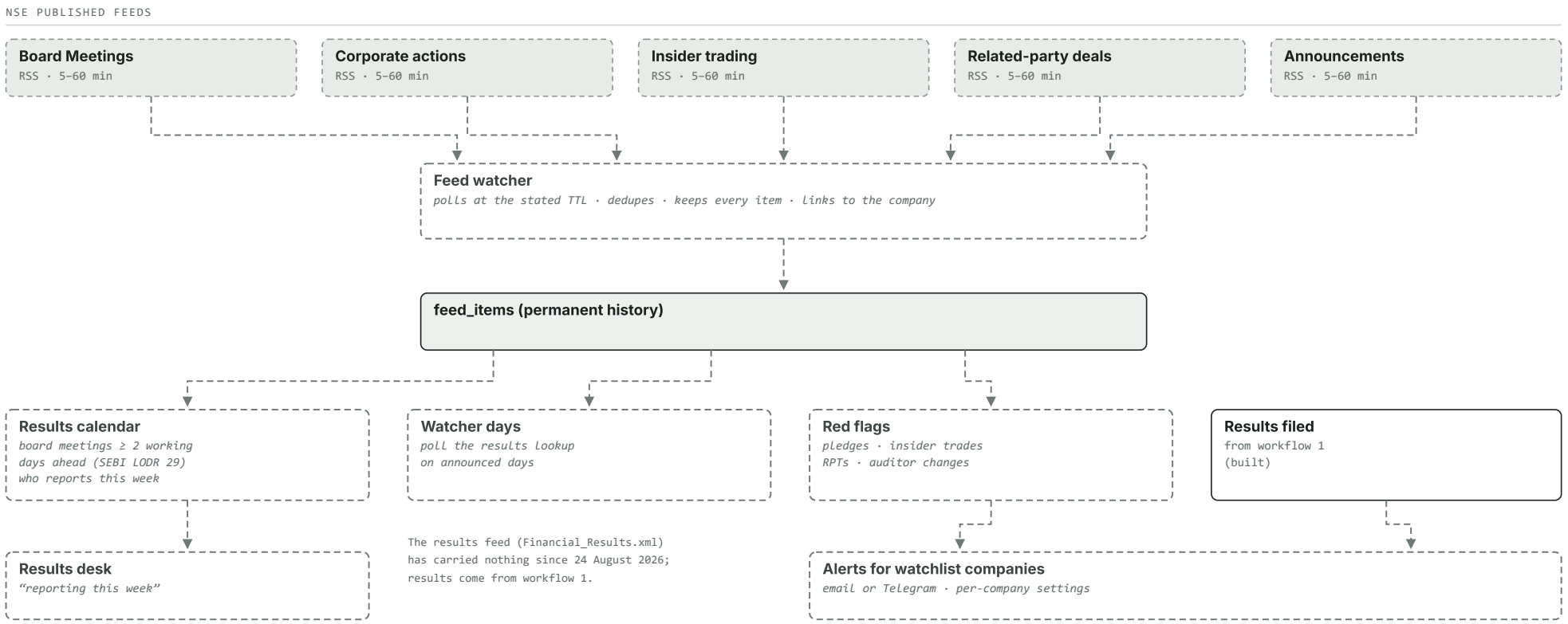
Waiting on: a valid Anthropic API key, a presentation and transcript source, and the embeddings provider (hosted or local).

Needs an always-on machine during the call (section 22.7); phone-only calls would need a paid telephony service. Transcripts stay in the analyst's own database; XBRL wins on any figure.

Only selected pages reach Claude, and every number is checked. For watchlist or on-demand companies, PDFs are read locally, pages classified and only the relevant ones sent for extraction against a JSON schema, each value with its page and quote. XBRL wins over any AI value. Drafts are written in a tool loop over the database and the model; a grounding check blocks any number that did not come from a tool result or a calculation. Future scope: a listener that joins the earnings call as it happens and transcribes it live, until the filed recording replaces it (section 26.3).

Workflow 9 · Events, results calendar and alerts

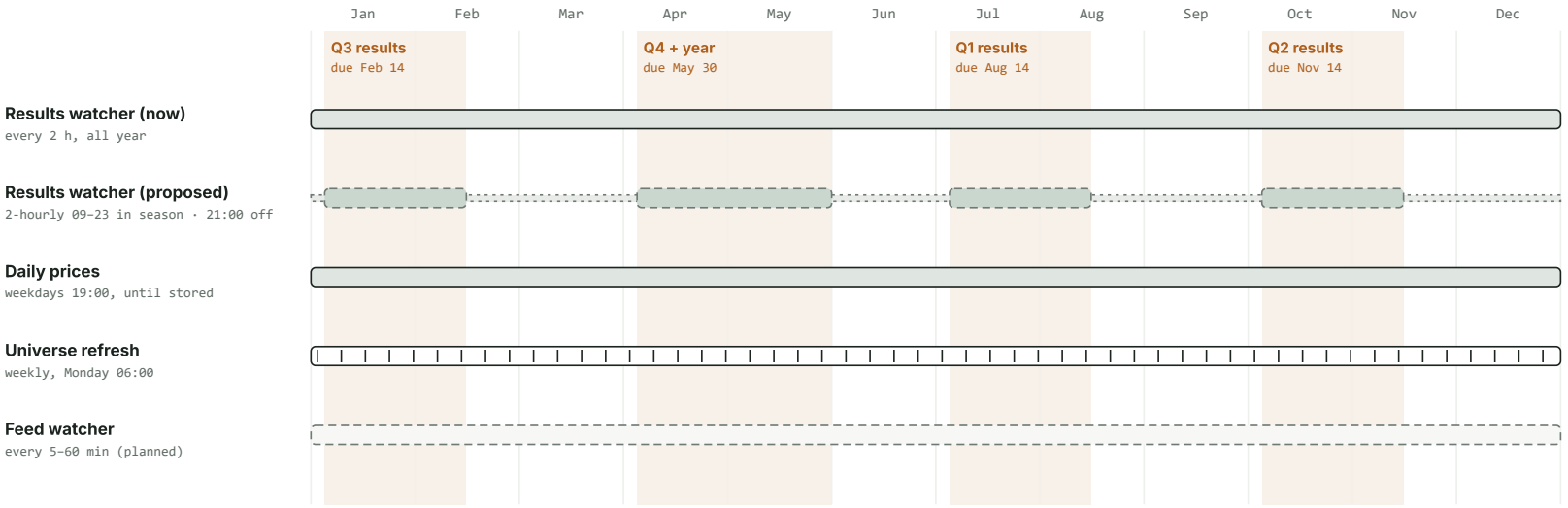
PLANNED · PHASES 1 AND 5



NSE’s published event feeds. Companies must announce a results board meeting at least two working days ahead (SEBI LODR 29), so the Board Meetings feed can show who reports this week and tell the results watcher which days matter. Watchlist companies get alerts when they file results or when a red flag appears (pledges, insider trades, related-party deals, auditor changes).

When each job runs

BUILT SEASON SCHEDULE PROPOSED

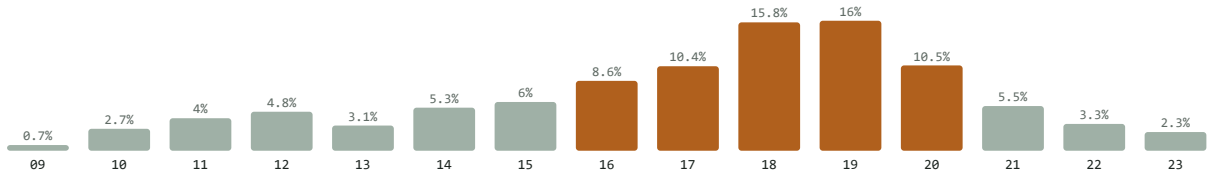


Requests a year for the results watcher: about 4,400 now (12 a day) → about 1,650 proposed. The windows run from day 5 to two days past each deadline.

When results are filed (IST)

share of filings by hour; 00–09 h: 1.0%

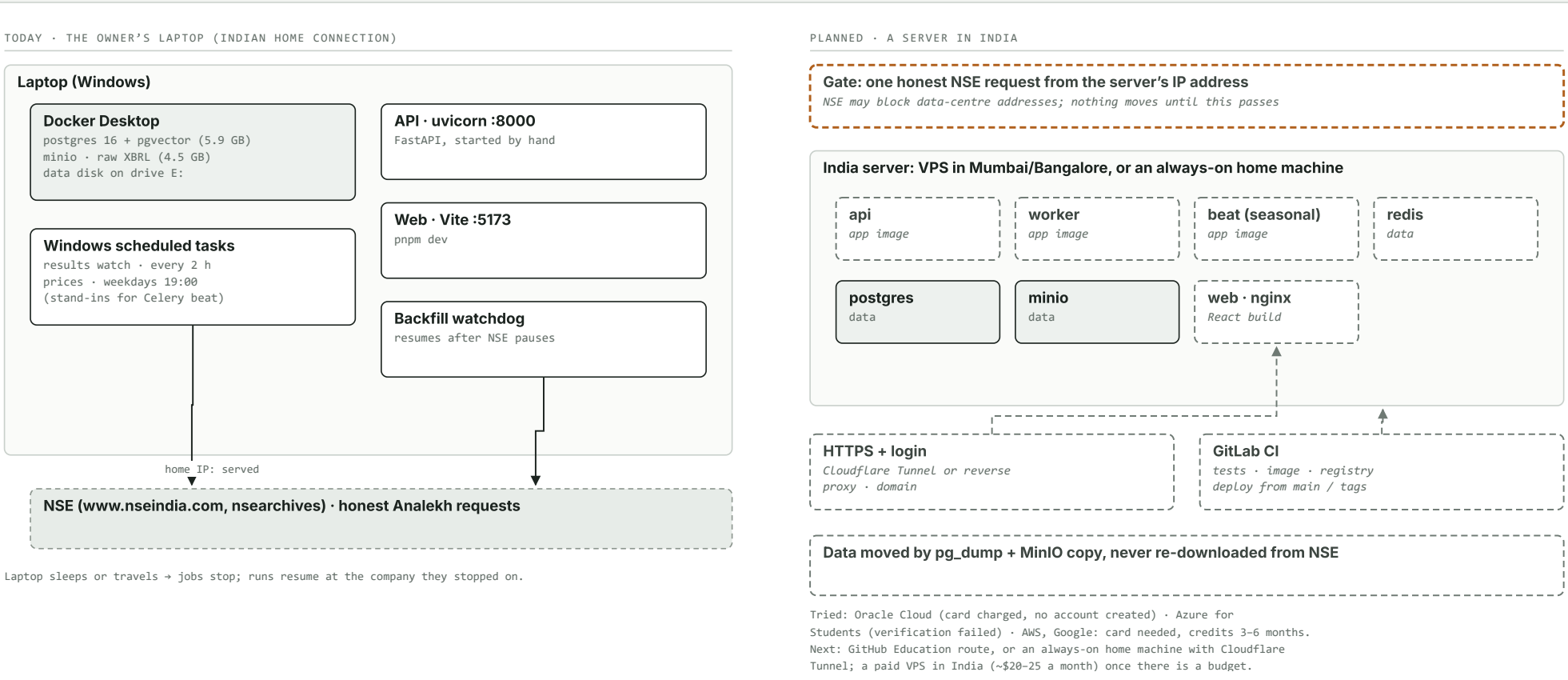
16–20 h: 61%



Results arrive in four seasons. SEBI requires quarterly results within 45 days of the quarter (60 for the year end). NSE showed integrated filings a median of one second after the company filed (299 filings, 24 companies), so any delay is the watcher's interval. Since 2023, every first filing of those companies fell between day 8 and day 59, inside the proposed windows; under 1% were filed between midnight and 09:00. The proposed schedule polls every 2 hours from 09:00 to 23:00 in season and once a day at 21:00 otherwise: about 60% fewer requests.

Where it runs

TODAY · PLANNED HOSTING



Numbers come from data

Figures in drafts are fetched from the database or the model through tool calls, and each one carries its XBRL tag or PDF page reference.

Data rights

NSE's published RSS feeds and lists, two lookups (owner-approved; per company, plus one request every 2 hours), one bhavcopy a day, NSE Indices' constituent CSVs. Requests identify as Analekh, never as a browser. Written confirmation or a licence before any public release.

Stack

Python 3.12 · FastAPI · Celery + Redis · lxml · PyMuPDF · Claude API · Postgres + pgvector · MinIO · xlwings · React + pnpm · Docker Compose · GitLab CI